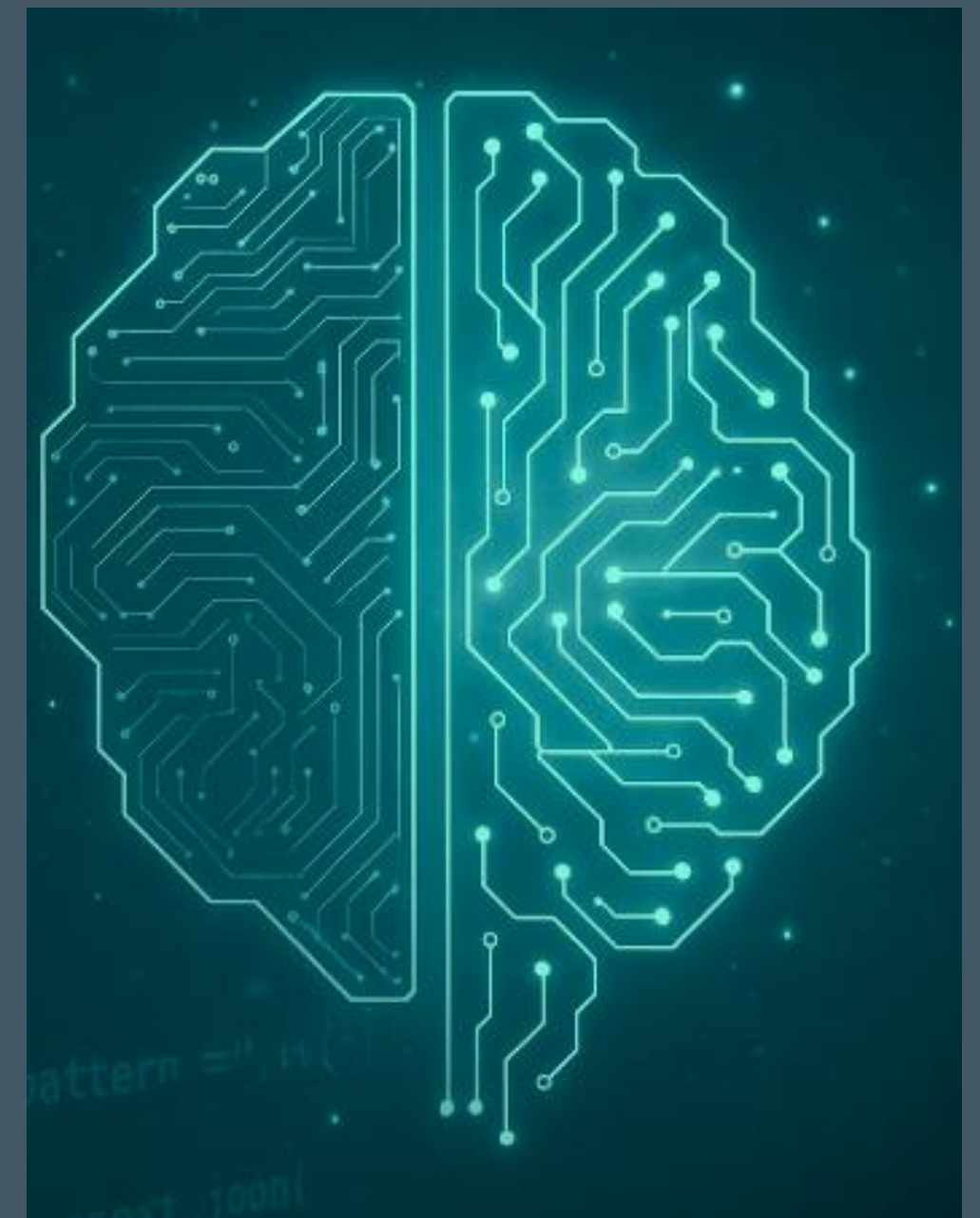

Title: Detection of AI-Generated Arabic Text

Advanced Data Analytic Techniques

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Problem Statement

- Arabic AI-text detection is still underdeveloped compared to English.
- Large Language Models (LLMs) like GPT-4, LLaMA, Jais can generate Arabic text almost identical to human writing.
- The lack of annotated Arabic datasets makes it difficult to detect AI-generated content.
- This raises risks in academic integrity, fake research, and digital misinformation.



Preprocessing & Data Normalization



A multi-stage preprocessing pipeline was implemented to handle the linguistic complexity of Arabic:

- **Text Normalization:**
 - Unify letter forms (ه → ة, ي → ى, ا → آ / إ / أ).
 - Remove diacritics (tashkeel), digits, and special symbols.
- **Structural Cleaning:**
 - Remove HTML tags, URLs, and email addresses.
 - Delete repeated punctuation and character elongations.
 - Normalize whitespace.

Preprocessing & Data Normalization Con't



- **Tokenization & Stopword Removal:**
 - Used NLTK Arabic tokenizer and extended stopwords lists (NLTK + GitHub custom).
- **Morphological Processing:**
 - ISRI stemming to reduce words to root forms.
 - Stanza lemmatization to preserve grammatical meaning.
- **Quality Assurance:**
 - Logs were used to track preprocessing errors.

Feature Engineering



1. Short Word Ratio (≤ 3 letters)

- AI:
 - Jais: 10.8% (highest – frequent short connectors).
 - Allam: 9.7%, LLaMA: 8.9%, OpenAI: 8.1%.
- Original: 9.8% (natural use of short function words).

Insight: Human and Jais texts show richer linguistic flow; OpenAI more formal and structured.

Feature Engineering Con't



2. Sentence Count & Structure

- AI:
- Allam: 4.49, Jais: 5.91, LLaMA: 5.66, OpenAI: 6.50 sentences on average.
- Original: 7.43 sentences (longer and more complex).

Insight: Human texts are more segmented and syntactically rich; AI outputs are concise and dense.

3. Foreign Letters Ratio

- AI: LLaMA: 0.98, Allam & Jais: 0.49, OpenAI: 0.29.
- Original: 0.83 (includes real English abbreviations or citations).

Insight: Human writing often mixes Arabic with foreign symbols; AI-generated text remains purely Arabic.

Feature Engineering Con't



4. Redundancy Score

- **AI: Allam: 0.56, LLaMA: 0.55, Jais: 0.54, OpenAI: 0.52.**
- **Original: 0.39** (much lower repetition).

Insight: AI-generated abstracts are more repetitive and template-like; human writing is diverse and context-driven.

5. Vocabulary Richness (TTR)

- **AI: 0.70–0.79** range.
- **Original: 0.798** (highest diversity).

Insight: Human abstracts use a broader and more varied vocabulary.

Embeddings & Data Splitting



TF-IDF (Term Frequency–Inverse Document Frequency)

- Used to represent texts numerically based on word frequency.
- Captures lexical patterns and stylistic differences—ideal for classical ML models.

BERT Embeddings (Contextual Representation)

- Transforms text into dense vectors capturing meaning and context.
- Enables deep models to understand semantics beyond surface words.

Data Splitting (70 / 15 / 15 : Train / Validation / Test)

- Ensures fair model evaluation and prevents data leakage.
- Training used for fitting, validation for tuning, and test for final performance check.

```
SUMMARY
=====
      Name  Type      Shape
_X_train dense (46944, 384)
_y_train dense   (46944,)
RT_X_val dense (5479, 384)
RT_y_val dense   (5479,)
T_X_test dense (5479, 384)
T_y_test dense   (5479,)
```

```
OF SUMMARY
=====
      Name      Type
OF_X_train sparse csr_matrix
OF_y_train      dense
IDF_X_val sparse csr_matrix
IDF_y_val      dense
IDF_X_test sparse csr_matrix
IDF_y_test      dense
-----
```

Data Balancing & Feature Scaling



SMOTE:

- Balanced the dataset by generating synthetic samples for the minority (human-written) class to prevent model bias.

StandardScaler:

- Normalized numerical features (zero mean, unit variance) to improve model convergence and performance consistency.

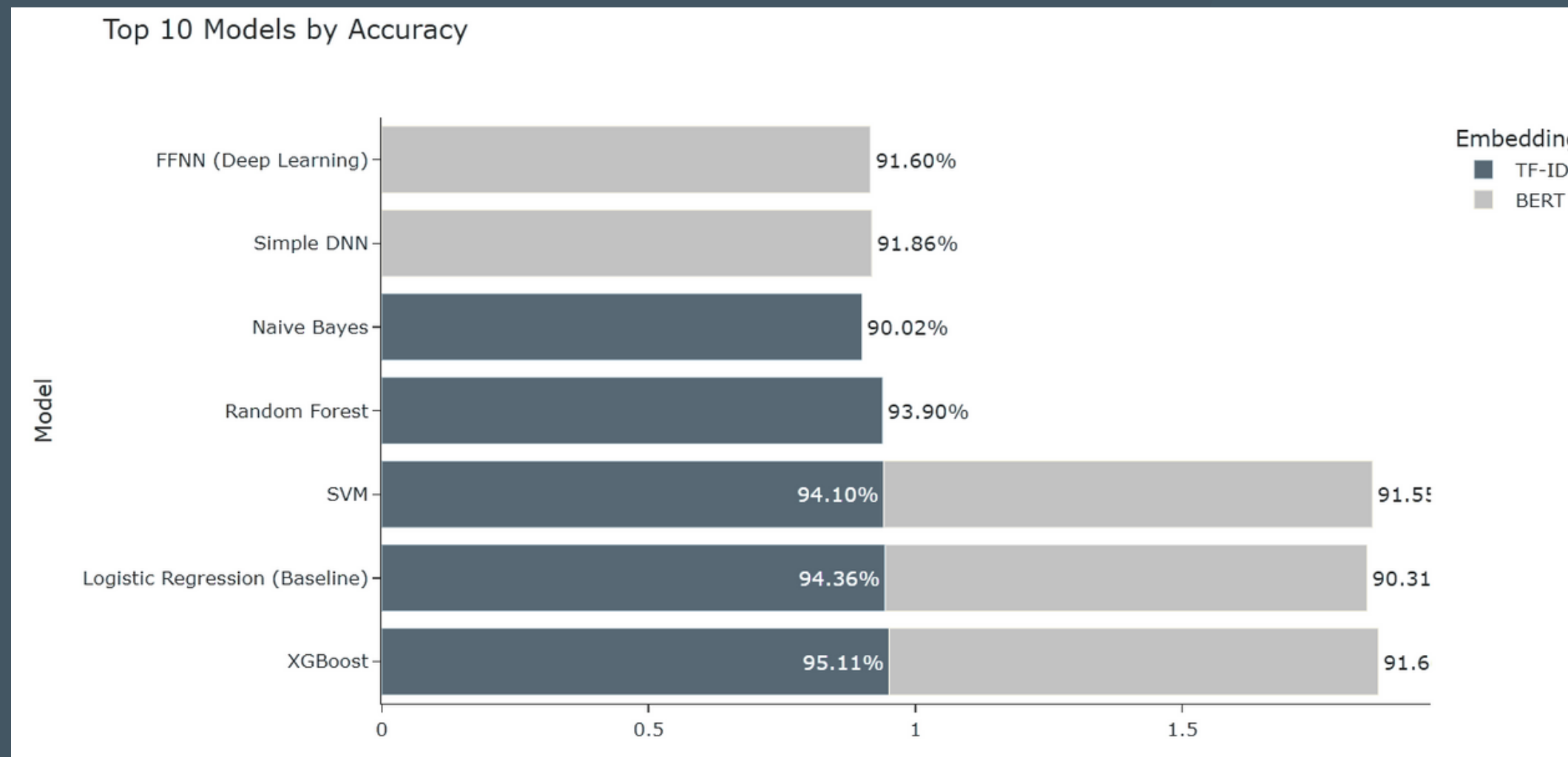
```
Class distribution:
train : {0: 23472, 1: 23472}
val    : {0: 5030, 1: 449}
test   : {0: 5030, 1: 449}
```

```
=====
```

Results & Analysis



- TF-IDF features were more effective for traditional ML models.
- XGBoost provided the best balance between accuracy and interpretability.
- BERT embeddings worked better for deep models, improving contextual generalization.
- Stylometric analysis validated these results — AI-generated texts were shorter, more redundant, and structurally repetitive compared to human texts.



Conclusion

- Developed a complete Arabic AI-text detection pipeline — from data collection to model evaluation.
 - Built a custom Arabic preprocessing framework with normalization, diacritics removal, stemming, and lemmatization (Stanza + ISRI).
 - Extracted stylometric features such as sentence count, redundancy score, short-word ratio, and vocabulary richness to capture writing patterns.
 - Applied SMOTE to balance the dataset and StandardScaler to normalize numerical features, ensuring fair model training and consistent performance.
 - Compared TF-IDF (lexical) and BERT (contextual) embeddings across classical and deep learning models.
 - XGBoost with TF-IDF achieved the best overall performance (95.1% accuracy, 0.950 F1-score), confirming the strength of lexical features for Arabic text classification.
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Thanks